

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 188 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 188

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

Your startup has a web application that currently serves customers out of a single region in Asia. You are targeting funding that will allow your startup to serve customers globally. Your current goal is to optimize for cost, and your post-funding goal is to optimize for global presence and performance. You must use a native JDBC driver. What should you do?

- A. Use Cloud Spanner to configure a single region instance initially, and then configure multi-region Cloud Spanner instances after securing funding. **Most Voted**
- B. Use a Cloud SQL for PostgreSQL highly available instance first, and Bigtable with US, Europe, and Asia replication after securing funding.
- C. Use a Cloud SQL for PostgreSQL zonal instance first, and Bigtable with US, Europe, and Asia after securing funding.
- D. Use a Cloud SQL for PostgreSQL zonal instance first, and Cloud SQL for PostgreSQL with highly available configuration after securing funding.

Hide Answer

Suggested Answer: C 

Community vote distribution

A (75%)

D (25%)

by  AWSandeep at Sept. 2, 2022, 11:06 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 136 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 136

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You are running a pipeline in Dataflow that receives messages from a Pub/Sub topic and writes the results to a BigQuery dataset in the EU. Currently, your pipeline is located in europe-west4 and has a maximum of 3 workers, instance type n1-standard-1. You notice that during peak periods, your pipeline is struggling to process records in a timely fashion, when all 3 workers are at maximum CPU utilization. Which two actions can you take to increase performance of your pipeline? (Choose two.)

A. Increase the number of max workers **Most Voted**

B. Use a larger instance type for your Dataflow workers **Most Voted**

C. Change the zone of your Dataflow pipeline to run in us-central1

D. Create a temporary table in Bigtable that will act as a buffer for new data. Create a new step in your pipeline to write to this table first, and then create a new pipeline to write from Bigtable to BigQuery

E. Create a temporary table in Cloud Spanner that will act as a buffer for new data. Create a new step in your pipeline to write to this table first, and then create a new pipeline to write from Cloud Spanner to BigQuery

Hide Answer

Suggested Answer: BE 

Community vote distribution

11 votes.

AB (100%)

by [jvg637](#) at March 18, 2020, 4:39 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 119 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 119

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You operate a database that stores stock trades and an application that retrieves average stock price for a given company over an adjustable window of time. The data is stored in Cloud Bigtable where the datetime of the stock trade is the beginning of the row key. Your application has thousands of concurrent users, and you notice that performance is starting to degrade as more stocks are added. What should you do to improve the performance of your application?

- A. Change the row key syntax in your Cloud Bigtable table to begin with the stock symbol. **Most Voted**
- B. Change the row key syntax in your Cloud Bigtable table to begin with a random number per second.
- C. Change the data pipeline to use BigQuery for storing stock trades, and update your application.
- D. Use Cloud Dataflow to write a summary of each day's stock trades to an Avro file on Cloud Storage. Update your application to read from Cloud Storage and Cloud Bigtable to compute the responses.

Hide Answer

Suggested Answer: A 

Community vote distribution

A (100%)

by [deleted] at March 22, 2020, 12:41 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 179 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 179

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You are building a real-time prediction engine that streams files, which may contain PII (personal identifiable information) data, into Cloud Storage and eventually into BigQuery. You want to ensure that the sensitive data is masked but still maintains referential integrity, because names and emails are often used as join keys. How should you use the Cloud Data Loss Prevention API (DLP API) to ensure that the PII data is not accessible by unauthorized individuals?

- A. Create a pseudonym by replacing the PII data with cryptogenic tokens, and store the non-tokenized data in a locked-down bucket. **Most Voted**
- B. Redact all PII data, and store a version of the unredacted data in a locked-down bucket.
- C. Scan every table in BigQuery, and mask the data it finds that has PII.
- D. Create a pseudonym by replacing PII data with a cryptographic format-preserving token. **Most Voted**

Hide Answer

Suggested Answer: B 

Community vote distribution

D (58%)

A (42%)

by  AWSandeeep at Sept. 2, 2022, 9:18 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 203 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 203

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

A TensorFlow machine learning model on Compute Engine virtual machines (n2-standard-32) takes two days to complete training. The model has custom TensorFlow operations that must run partially on a CPU. You want to reduce the training time in a cost-effective manner. What should you do?

- A. Change the VM type to n2-highmem-32.
- B. Change the VM type to e2-standard-32.
- C. Train the model using a VM with a GPU hardware accelerator.
- D. Train the model using a VM with a TPU hardware accelerator.

Hide Answer

Suggested Answer: C 

Community vote distribution

C (100%)

by  gudiking at Nov. 29, 2022, 2:15 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 154 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 154

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You plan to deploy Cloud SQL using MySQL. You need to ensure high availability in the event of a zone failure. What should you do?

- A. Create a Cloud SQL instance in one zone, and create a failover replica in another zone within the same region. **Most Voted**
- B. Create a Cloud SQL instance in one zone, and create a read replica in another zone within the same region.
- C. Create a Cloud SQL instance in one zone, and configure an external read replica in a zone in a different region.
- D. Create a Cloud SQL instance in a region, and configure automatic backup to a Cloud Storage bucket in the same region.

Hide Answer

Suggested Answer: C 

Community vote distribution

A (88%)

13%

by  madhu1171 at March 15, 2020, 7:17 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 118 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 118

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You need to set access to BigQuery for different departments within your company. Your solution should comply with the following requirements:

☞ Each department should have access only to their data.

Each department will have one or more leads who need to be able to create and update tables and provide them to their team.

☞ Each department has data analysts who need to be able to query but not modify data.

How should you set access to the data in BigQuery?

A. Create a dataset for each department. Assign the department leads the role of OWNER, and assign the data analysts the role of WRITER on their dataset.

B. Create a dataset for each department. Assign the department leads the role of WRITER, and assign the data analysts the role of READER on their dataset.

Most Voted

C. Create a table for each department. Assign the department leads the role of Owner, and assign the data analysts the role of Editor on the project the table is in.

D. Create a table for each department. Assign the department leads the role of Editor, and assign the data analysts the role of Viewer on the project the table is in.

Hide Answer

Suggested Answer: D 

Community vote distribution

B (100%)

by  jvg637 at March 18, 2020, 8:49 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 173 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 173

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You are designing a pipeline that publishes application events to a Pub/Sub topic. Although message ordering is not important, you need to be able to aggregate events across disjoint hourly intervals before loading the results to BigQuery for analysis. What technology should you use to process and load this data to BigQuery while ensuring that it will scale with large volumes of events?

- A. Create a Cloud Function to perform the necessary data processing that executes using the Pub/Sub trigger every time a new message is published to the topic.
- B. Schedule a Cloud Function to run hourly, pulling all available messages from the Pub/Sub topic and performing the necessary aggregations.
- C. Schedule a batch Dataflow job to run hourly, pulling all available messages from the Pub/Sub topic and performing the necessary aggregations.
- D. Create a streaming Dataflow job that reads continually from the Pub/Sub topic and performs the necessary aggregations using tumbling windows. **Most Voted**

Hide Answer

Suggested Answer: D 

Community vote distribution

D (100%)

by  AWSandeep at Sept. 2, 2022, 7:51 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 191 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 191

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You are developing a new deep learning model that predicts a customer's likelihood to buy on your ecommerce site. After running an evaluation of the model against both the original training data and new test data, you find that your model is overfitting the data. You want to improve the accuracy of the model when predicting new data. What should you do?

- A. Increase the size of the training dataset, and increase the number of input features.
- B. Increase the size of the training dataset, and decrease the number of input features. **Most Voted**
- C. Reduce the size of the training dataset, and increase the number of input features.
- D. Reduce the size of the training dataset, and decrease the number of input features.

Hide Answer

Suggested Answer: A 

Community vote distribution

B (92%)

8%

by  ducc at Sept. 3, 2022, 3:46 a.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 164 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 164

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You are working on a linear regression model on BigQuery ML to predict a customer's likelihood of purchasing your company's products. Your model uses a city name variable as a key predictive component. In order to train and serve the model, your data must be organized in columns. You want to prepare your data using the least amount of coding while maintaining the predictable variables. What should you do?

- A. Create a new view with BigQuery that does not include a column with city information.
- B. Use SQL in BigQuery to transform the state column using a one-hot encoding method, and make each city a column with binary values. **Most Voted**
- C. Use TensorFlow to create a categorical variable with a vocabulary list. Create the vocabulary file and upload that as part of your model to BigQuery ML.
- D. Use Cloud Data Fusion to assign each city to a region that is labeled as 1, 2, 3, 4, or 5, and then use that number to represent the city in the model. **Most Voted**

Hide Answer

Suggested Answer: D 

Community vote distribution

B (67%)

D (33%)

by [AWSandeep](#) at Sept. 2, 2022, 6:25 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 115 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 115

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You use a dataset in BigQuery for analysis. You want to provide third-party companies with access to the same dataset. You need to keep the costs of data sharing low and ensure that the data is current. Which solution should you choose?

- A. Create an authorized view on the BigQuery table to control data access, and provide third-party companies with access to that view. **Most Voted**
- B. Use Cloud Scheduler to export the data on a regular basis to Cloud Storage, and provide third-party companies with access to the bucket.
- C. Create a separate dataset in BigQuery that contains the relevant data to share, and provide third-party companies with access to the new dataset.
- D. Create a Cloud Dataflow job that reads the data in frequent time intervals, and writes it to the relevant BigQuery dataset or Cloud Storage bucket for third-party companies to use.

Hide Answer

Suggested Answer: B 

Community vote distribution

A (100%)

by [jvg637](#) at March 19, 2020, 10:20 a.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 128 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 128

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You work on a regression problem in a natural language processing domain, and you have 100M labeled examples in your dataset. You have randomly shuffled your data and split your dataset into train and test samples (in a 90/10 ratio). After you trained the neural network and evaluated your model on a test set, you discover that the root-mean-squared error (RMSE) of your model is twice as high on the train set as on the test set. How should you improve the performance of your model?

- A. Increase the share of the test sample in the train-test split.
- B. Try to collect more data and increase the size of your dataset.
- C. Try out regularization techniques (e.g., dropout or batch normalization) to avoid overfitting.
- D. Increase the complexity of your model by, e.g., introducing an additional layer or increase sizing the size of vocabularies or n-grams used. **Most Voted**

Hide Answer

Suggested Answer: D 

Community vote distribution

D (88%)

13%

by [deleted] at March 22, 2020, 11:11 a.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 102 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 102

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You need to create a near real-time inventory dashboard that reads the main inventory tables in your BigQuery data warehouse. Historical inventory data is stored as inventory balances by item and location. You have several thousand updates to inventory every hour. You want to maximize performance of the dashboard and ensure that the data is accurate. What should you do?

- A. Leverage BigQuery UPDATE statements to update the inventory balances as they are changing. **Most Voted**
- B. Partition the inventory balance table by item to reduce the amount of data scanned with each inventory update.
- C. Use the BigQuery streaming the stream changes into a daily inventory movement table. Calculate balances in a view that joins it to the historical inventory balance table. Update the inventory balance table nightly.
- D. Use the BigQuery bulk loader to batch load inventory changes into a daily inventory movement table. Calculate balances in a view that joins it to the historical inventory balance table. Update the inventory balance table nightly.

Hide Answer

Suggested Answer: C 

Community vote distribution

A (57%)

C (43%)

by  [Rajokkiyam](#) at March 22, 2020, 6:37 a.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 106 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 106

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You are managing a Cloud Dataproc cluster. You need to make a job run faster while minimizing costs, without losing work in progress on your clusters. What should you do?

- A. Increase the cluster size with more non-preemptible workers.
- B. Increase the cluster size with preemptible worker nodes, and configure them to forcefully decommission.
- C. Increase the cluster size with preemptible worker nodes, and use Cloud Stackdriver to trigger a script to preserve work.
- D. Increase the cluster size with preemptible worker nodes, and configure them to use graceful decommissioning.

Hide Answer

Suggested Answer: D 

Reference -

<https://cloud.google.com/dataproc/docs/concepts/configuring-clusters/flex>

Community vote distribution

D (67%)

A (33%)

by  [Rajokkiyam](#) at March 22, 2020, 6:42 a.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 120 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 120

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You are operating a Cloud Dataflow streaming pipeline. The pipeline aggregates events from a Cloud Pub/Sub subscription source, within a window, and sinks the resulting aggregation to a Cloud Storage bucket. The source has consistent throughput. You want to monitor an alert on behavior of the pipeline with Cloud Stackdriver to ensure that it is processing data. Which Stackdriver alerts should you create?

- A. An alert based on a decrease of subscription/num_undelivered_messages for the source and a rate of change increase of instance/storage/ used_bytes for the destination
- B. An alert based on an increase of subscription/num_undelivered_messages for the source and a rate of change decrease of instance/storage/ used_bytes for the destination
- C. An alert based on a decrease of instance/storage/used_bytes for the source and a rate of change increase of subscription/ num_undelivered_messages for the destination
- D. An alert based on an increase of instance/storage/used_bytes for the source and a rate of change decrease of subscription/ num_undelivered_messages for the destination

Hide Answer

Suggested Answer: B 

Community vote distribution

B (80%)

A (20%)

by [deleted] at March 22, 2020, 12:46 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 134 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 134

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You are building an application to share financial market data with consumers, who will receive data feeds. Data is collected from the markets in real time. Consumers will receive the data in the following ways:

- ☐ Real-time event stream
- ☐ ANSI SQL access to real-time stream and historical data
- ☐ Batch historical exports

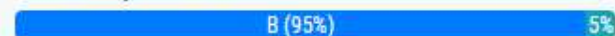
Which solution should you use?

- A. Cloud Dataflow, Cloud SQL, Cloud Spanner
- B. Cloud Pub/Sub, Cloud Storage, BigQuery **Most Voted**
- C. Cloud Dataproc, Cloud Dataflow, BigQuery
- D. Cloud Pub/Sub, Cloud Dataproc, Cloud SQL

Hide Answer

Suggested Answer: A 

Community vote distribution



by [deleted] at March 22, 2020, 10:43 a.m.

EXAM PROFESSIONAL MACHINE LEARNING ENGINEER TOPIC 1 QUESTION 4 DISCUSSION

Actual exam question from Google's Professional Machine Learning Engineer

Question #: 4

Topic #: 1

[\[All Professional Machine Learning Engineer Questions\]](#)

You want to rebuild your ML pipeline for structured data on Google Cloud. You are using PySpark to conduct data transformations at scale, but your pipelines are taking over 12 hours to run. To speed up development and pipeline run time, you want to use a serverless tool and SQL syntax. You have already moved your raw data into Cloud Storage. How should you build the pipeline on Google Cloud while meeting the speed and processing requirements?

- A. Use Data Fusion's GUI to build the transformation pipelines, and then write the data into BigQuery.
- B. Convert your PySpark into SparkSQL queries to transform the data, and then run your pipeline on Dataproc to write the data into BigQuery.
- C. Ingest your data into Cloud SQL, convert your PySpark commands into SQL queries to transform the data, and then use federated queries from BigQuery for machine learning.
- D. Ingest your data into BigQuery using BigQuery Load, convert your PySpark commands into BigQuery SQL queries to transform the data, and then write the transformations to a new table. **Most Voted**

Hide Answer

Suggested Answer: B 

Community vote distribution

D (69%)

A (15%)

B (15%)

by  ralf_cc at July 8, 2021, 12:35 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 131 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 131

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

As your organization expands its usage of GCP, many teams have started to create their own projects. Projects are further multiplied to accommodate different stages of deployments and target audiences. Each project requires unique access control configurations. The central IT team needs to have access to all projects. Furthermore, data from Cloud Storage buckets and BigQuery datasets must be shared for use in other projects in an ad hoc way. You want to simplify access control management by minimizing the number of policies. Which two steps should you take? (Choose two.)

A. Use Cloud Deployment Manager to automate access provision.

B. Introduce resource hierarchy to leverage access control policy inheritance. **Most Voted**

C. Create distinct groups for various teams, and specify groups in Cloud IAM policies. **Most Voted**

D. Only use service accounts when sharing data for Cloud Storage buckets and BigQuery datasets.

E. For each Cloud Storage bucket or BigQuery dataset, decide which projects need access. Find all the active members who have access to these projects, and create a Cloud IAM policy to grant access to all these users.

Hide Answer

Suggested Answer: AC 

Community vote distribution

BC (76%)

AC (24%)

by [deleted] at March 22, 2020, 10:12 a.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 115 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 115

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You use a dataset in BigQuery for analysis. You want to provide third-party companies with access to the same dataset. You need to keep the costs of data sharing low and ensure that the data is current. Which solution should you choose?

- A. Create an authorized view on the BigQuery table to control data access, and provide third-party companies with access to that view. **Most Voted**
- B. Use Cloud Scheduler to export the data on a regular basis to Cloud Storage, and provide third-party companies with access to the bucket.
- C. Create a separate dataset in BigQuery that contains the relevant data to share, and provide third-party companies with access to the new dataset.
- D. Create a Cloud Dataflow job that reads the data in frequent time intervals, and writes it to the relevant BigQuery dataset or Cloud Storage bucket for third-party companies to use.

Hide Answer

Suggested Answer: B 

Community vote distribution

A (100%)

by [jvg637](#) at March 19, 2020, 10:20 a.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 156 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 156

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You are planning to migrate your current on-premises Apache Hadoop deployment to the cloud. You need to ensure that the deployment is as fault-tolerant and cost-effective as possible for long-running batch jobs. You want to use a managed service. What should you do?

- A. Deploy a Dataproc cluster. Use a standard persistent disk and 50% preemptible workers. Store data in Cloud Storage, and change references in scripts from `hdfs://` to `gs://`
- B. Deploy a Dataproc cluster. Use an SSD persistent disk and 50% preemptible workers. Store data in Cloud Storage, and change references in scripts from `hdfs://` to `gs://`
- C. Install Hadoop and Spark on a 10-node Compute Engine instance group with standard instances. Install the Cloud Storage connector, and store the data in Cloud Storage. Change references in scripts from `hdfs://` to `gs://`
- D. Install Hadoop and Spark on a 10-node Compute Engine instance group with preemptible instances. Store data in HDFS. Change references in scripts from `hdfs://` to `gs://`

Hide Answer

Suggested Answer: A 

Community vote distribution

A (100%)

by [deleted] at March 22, 2020, 7:31 a.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 137 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 137

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You have a data pipeline with a Cloud Dataflow job that aggregates and writes time series metrics to Cloud Bigtable. This data feeds a dashboard used by thousands of users across the organization. You need to support additional concurrent users and reduce the amount of time required to write the data. Which two actions should you take? (Choose two.)

- A. Configure your Cloud Dataflow pipeline to use local execution
- B. Increase the maximum number of Cloud Dataflow workers by setting `maxNumWorkers` in `PipelineOptions` **Most Voted**
- C. Increase the number of nodes in the Cloud Bigtable cluster **Most Voted**
- D. Modify your Cloud Dataflow pipeline to use the Flatten transform before writing to Cloud Bigtable
- E. Modify your Cloud Dataflow pipeline to use the CoGroupByKey transform before writing to Cloud Bigtable

Hide Answer

Suggested Answer: DE 

Reference:

<https://www.slideshare.net/LucasArruda3/how-to-build-an-etl-pipeline-with-apache-beam-on-google-cloud-dataflow>

Community vote distribution



by  rickywck at March 17, 2020, 1:50 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 141 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 141

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

Data Analysts in your company have the Cloud IAM Owner role assigned to them in their projects to allow them to work with multiple GCP products in their projects. Your organization requires that all BigQuery data access logs be retained for 6 months. You need to ensure that only audit personnel in your company can access the data access logs for all projects. What should you do?

- A. Enable data access logs in each Data Analyst's project. Restrict access to Stackdriver Logging via Cloud IAM roles.
- B. Export the data access logs via a project-level export sink to a Cloud Storage bucket in the Data Analysts' projects. Restrict access to the Cloud Storage bucket.
- C. Export the data access logs via a project-level export sink to a Cloud Storage bucket in a newly created projects for audit logs. Restrict access to the project with the exported logs.
- D. Export the data access logs via an aggregated export sink to a Cloud Storage bucket in a newly created project for audit logs. Restrict access to the project that contains the exported logs. **Most Voted**

Hide Answer

Suggested Answer: D 

Community vote distribution

D (100%)

by [deleted] at March 22, 2020, 8:58 a.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 197 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 197

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You are designing a system that requires an ACID-compliant database. You must ensure that the system requires minimal human intervention in case of a failure. What should you do?

- A. Configure a Cloud SQL for MySQL instance with point-in-time recovery enabled.
- B. Configure a Cloud SQL for PostgreSQL instance with high availability enabled. **Most Voted**
- C. Configure a Bigtable instance with more than one cluster.
- D. Configure a BigQuery table with a multi-region configuration.

Hide Answer

Suggested Answer: B 

Community vote distribution

B (100%)

by  **ducc** at Sept. 3, 2022, 3:57 a.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 203 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 203

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

A TensorFlow machine learning model on Compute Engine virtual machines (n2-standard-32) takes two days to complete training. The model has custom TensorFlow operations that must run partially on a CPU. You want to reduce the training time in a cost-effective manner. What should you do?

- A. Change the VM type to n2-highmem-32.
- B. Change the VM type to e2-standard-32.
- C. Train the model using a VM with a GPU hardware accelerator.
- D. Train the model using a VM with a TPU hardware accelerator.

Hide Answer

Suggested Answer: C 

Community vote distribution

C (100%)

by  gudiking at Nov. 29, 2022, 2:15 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 205 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 205

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You have a data processing application that runs on Google Kubernetes Engine (GKE). Containers need to be launched with their latest available configurations from a container registry. Your GKE nodes need to have GPUs, local SSDs, and 8 Gbps bandwidth. You want to efficiently provision the data processing infrastructure and manage the deployment process. What should you do?

- A. Use Compute Engine startup scripts to pull container images, and use gcloud commands to provision the infrastructure.
- B. Use Cloud Build to schedule a job using Terraform build to provision the infrastructure and launch with the most current container images. **Most Voted**
- C. Use GKE to autoscale containers, and use gcloud commands to provision the infrastructure.
- D. Use Dataflow to provision the data pipeline, and use Cloud Scheduler to run the job.

Hide Answer

Suggested Answer: C 

Community vote distribution

B (100%)

by  **Atnafu** at Nov. 30, 2022, 11:10 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 138 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 138

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You have several Spark jobs that run on a Cloud Dataproc cluster on a schedule. Some of the jobs run in sequence, and some of the jobs run concurrently. You need to automate this process. What should you do?

- A. Create a Cloud Dataproc Workflow Template
- B. Create an initialization action to execute the jobs
- C. Create a Directed Acyclic Graph in Cloud Composer **Most Voted**
- D. Create a Bash script that uses the Cloud SDK to create a cluster, execute jobs, and then tear down the cluster

Hide Answer

Suggested Answer: A 

Reference:

<https://cloud.google.com/dataproc/docs/concepts/workflows/using-workflows>

Community vote distribution

C (100%)

by  jvg637 at March 18, 2020, 5:08 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 94 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 94

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You are designing an Apache Beam pipeline to enrich data from Cloud Pub/Sub with static reference data from BigQuery. The reference data is small enough to fit in memory on a single worker. The pipeline should write enriched results to BigQuery for analysis. Which job type and transforms should this pipeline use?

- A. Batch job, PubSubIO, side-inputs
- B. Streaming job, PubSubIO, JdbcIO, side-outputs
- C. Streaming job, PubSubIO, BigQueryIO, side-inputs **Most Voted**
- D. Streaming job, PubSubIO, BigQueryIO, side-outputs

Hide Answer

Suggested Answer: C 

Community vote distribution

C (100%)

by  rickywck at March 17, 2020, 9:34 a.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 168 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 168

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You work for a financial institution that lets customers register online. As new customers register, their user data is sent to Pub/Sub before being ingested into BigQuery. For security reasons, you decide to redact your customers' Government issued Identification Number while allowing customer service representatives to view the original values when necessary. What should you do?

- A. Use BigQuery's built-in AEAD encryption to encrypt the SSN column. Save the keys to a new table that is only viewable by permissioned users.
- B. Use BigQuery column-level security. Set the table permissions so that only members of the Customer Service user group can see the SSN column. **Most Voted**
- C. Before loading the data into BigQuery, use Cloud Data Loss Prevention (DLP) to replace input values with a cryptographic hash.
- D. Before loading the data into BigQuery, use Cloud Data Loss Prevention (DLP) to replace input values with a cryptographic format-preserving encryption token. **Most Voted**

Hide Answer

Suggested Answer: *D* 

Community vote distribution



by  AWSandeep at Sept. 2, 2022, 7:01 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 181 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 181

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You need to give new website users a globally unique identifier (GUID) using a service that takes in data points and returns a GUID. This data is sourced from both internal and external systems via HTTP calls that you will make via microservices within your pipeline. There will be tens of thousands of messages per second and that can be multi-threaded, and you worry about the backpressure on the system. How should you design your pipeline to minimize that backpressure?

- A. Call out to the service via HTTP.
- B. Create the pipeline statically in the class definition.
- C. Create a new object in the startBundle method of DoFn.
- D. Batch the job into ten-second increments. **Most Voted**

Hide Answer

Suggested Answer: D 

Community vote distribution



by  ducc at Sept. 4, 2022, 3:19 a.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 110 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 110

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You are creating a new pipeline in Google Cloud to stream IoT data from Cloud Pub/Sub through Cloud Dataflow to BigQuery. While previewing the data, you notice that roughly 2% of the data appears to be corrupt. You need to modify the Cloud Dataflow pipeline to filter out this corrupt data. What should you do?

- A. Add a SideInput that returns a Boolean if the element is corrupt.
- B. Add a ParDo transform in Cloud Dataflow to discard corrupt elements. **Most Voted**
- C. Add a Partition transform in Cloud Dataflow to separate valid data from corrupt data.
- D. Add a GroupByKey transform in Cloud Dataflow to group all of the valid data together and discard the rest.

Hide Answer

Suggested Answer: B 

Community vote distribution

B (100%)

by [deleted] at March 22, 2020, 2:51 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 204 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 204

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You want to create a machine learning model using BigQuery ML and create an endpoint for hosting the model using Vertex AI. This will enable the processing of continuous streaming data in near-real time from multiple vendors. The data may contain invalid values. What should you do?

- A. Create a new BigQuery dataset and use streaming inserts to land the data from multiple vendors. Configure your BigQuery ML model to use the "ingestion" dataset as the framing data.
- B. Use BigQuery streaming inserts to land the data from multiple vendors where your BigQuery dataset ML model is deployed.
- C. Create a Pub/Sub topic and send all vendor data to it. Connect a Cloud Function to the topic to process the data and store it in BigQuery.
- D. Create a Pub/Sub topic and send all vendor data to it. Use Dataflow to process and sanitize the Pub/Sub data and stream it to BigQuery. **Most Voted**

Hide Answer

Suggested Answer: A 

Community vote distribution

D (100%)

by  Atnafu at Nov. 30, 2022, 11:08 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 150 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 150

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You want to build a managed Hadoop system as your data lake. The data transformation process is composed of a series of Hadoop jobs executed in sequence. To accomplish the design of separating storage from compute, you decided to use the Cloud Storage connector to store all input data, output data, and intermediary data. However, you noticed that one Hadoop job runs very slowly with Cloud Dataproc, when compared with the on-premises bare-metal Hadoop environment (8-core nodes with 100-GB RAM). Analysis shows that this particular Hadoop job is disk I/O intensive. You want to resolve the issue. What should you do?

- A. Allocate sufficient memory to the Hadoop cluster, so that the intermediary data of that particular Hadoop job can be held in memory
- B. Allocate sufficient persistent disk space to the Hadoop cluster, and store the intermediate data of that particular Hadoop job on native HDFS **Most Voted**
- C. Allocate more CPU cores of the virtual machine instances of the Hadoop cluster so that the networking bandwidth for each instance can scale up
- D. Allocate additional network interface card (NIC), and configure link aggregation in the operating system to use the combined throughput when working with Cloud Storage

Hide Answer

Suggested Answer: A 

Community vote distribution

B (100%)

by  rickywck at March 17, 2020, 2:58 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 201 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 201

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You need to migrate a Redis database from an on-premises data center to a Memorystore for Redis instance. You want to follow Google-recommended practices and perform the migration for minimal cost, time and effort. What should you do?

- A. Make an RDB backup of the Redis database, use the gsutil utility to copy the RDB file into a Cloud Storage bucket, and then import the RDB file into the Memorystore for Redis instance. **Most Voted**
- B. Make a secondary instance of the Redis database on a Compute Engine instance and then perform a live cutover.
- C. Create a Dataflow job to read the Redis database from the on-premises data center and write the data to a Memorystore for Redis instance.
- D. Write a shell script to migrate the Redis data and create a new Memorystore for Redis instance.

Hide Answer

Suggested Answer: B 

Community vote distribution

A (100%)

by  ducc at Sept. 3, 2022, 4:12 a.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 198 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 198

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You are implementing workflow pipeline scheduling using open source-based tools and Google Kubernetes Engine (GKE). You want to use a Google managed service to simplify and automate the task. You also want to accommodate Shared VPC networking considerations. What should you do?

- A. Use Dataflow for your workflow pipelines. Use Cloud Run triggers for scheduling.
- B. Use Dataflow for your workflow pipelines. Use shell scripts to schedule workflows.
- C. Use Cloud Composer in a Shared VPC configuration. Place the Cloud Composer resources in the host project.
- D. Use Cloud Composer in a Shared VPC configuration. Place the Cloud Composer resources in the service project. **Most Voted**

Hide Answer

Suggested Answer: A 

Community vote distribution

D (100%)

by  **ducc** at Sept. 3, 2022, 4 a.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 200 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 200

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

Government regulations in the banking industry mandate the protection of clients' personally identifiable information (PII). Your company requires PII to be access controlled, encrypted, and compliant with major data protection standards. In addition to using Cloud Data Loss Prevention (Cloud DLP), you want to follow Google-recommended practices and use service accounts to control access to PII. What should you do?

- A. Assign the required Identity and Access Management (IAM) roles to every employee, and create a single service account to access project resources. **Most Voted**
- B. Use one service account to access a Cloud SQL database, and use separate service accounts for each human user.
- C. Use Cloud Storage to comply with major data protection standards. Use one service account shared by all users.
- D. Use Cloud Storage to comply with major data protection standards. Use multiple service accounts attached to IAM groups to grant the appropriate access to each group. **Most Voted**

Hide Answer

Suggested Answer: C 

Community vote distribution

A (52%)

D (48%)

by  **ducc** at Sept. 3, 2022, 4:04 a.m.

EXAM PROFESSIONAL MACHINE LEARNING ENGINEER TOPIC 1 QUESTION 17 DISCUSSION

Actual exam question from Google's Professional Machine Learning Engineer

Question #: 17

Topic #: 1

[\[All Professional Machine Learning Engineer Questions\]](#)

You are building a real-time prediction engine that streams files which may contain Personally Identifiable Information (PII) to Google Cloud. You want to use the Cloud Data Loss Prevention (DLP) API to scan the files. How should you ensure that the PII is not accessible by unauthorized individuals?

- A. Stream all files to Google Cloud, and then write the data to BigQuery. Periodically conduct a bulk scan of the table using the DLP API.
- B. Stream all files to Google Cloud, and write batches of the data to BigQuery. While the data is being written to BigQuery, conduct a bulk scan of the data using the DLP API.
- C. Create two buckets of data: Sensitive and Non-sensitive. Write all data to the Non-sensitive bucket. Periodically conduct a bulk scan of that bucket using the DLP API, and move the sensitive data to the Sensitive bucket.
- D. Create three buckets of data: Quarantine, Sensitive, and Non-sensitive. Write all data to the Quarantine bucket. Periodically conduct a bulk scan of that bucket using the DLP API, and move the data to either the Sensitive or Non-Sensitive bucket. **Most Voted**

Hide Answer

Suggested Answer: A 

Community vote distribution



by  **chohan** at June 16, 2021, 1:58 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 123 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 123

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You need to create a data pipeline that copies time-series transaction data so that it can be queried from within BigQuery by your data science team for analysis. Every hour, thousands of transactions are updated with a new status. The size of the initial dataset is 1.5 PB, and it will grow by 3 TB per day. The data is heavily structured, and your data science team will build machine learning models based on this data. You want to maximize performance and usability for your data science team. Which two strategies should you adopt? (Choose two.)

- A. Denormalize the data as much as possible. **Most Voted**
- B. Preserve the structure of the data as much as possible.
- C. Use BigQuery UPDATE to further reduce the size of the dataset.
- D. Develop a data pipeline where status updates are appended to BigQuery instead of updated. **Most Voted**
- E. Copy a daily snapshot of transaction data to Cloud Storage and store it as an Avro file. Use BigQuery's support for external data sources to query.

Hide Answer

Suggested Answer: AD 

Community vote distribution

10 votes.

AD (91%)

9%

by  rickywck at March 20, 2020, 4:21 a.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 167 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 167

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

Your company currently runs a large on-premises cluster using Spark, Hive, and HDFS in a colocation facility. The cluster is designed to accommodate peak usage on the system; however, many jobs are batch in nature, and usage of the cluster fluctuates quite dramatically. Your company is eager to move to the cloud to reduce the overhead associated with on-premises infrastructure and maintenance and to benefit from the cost savings. They are also hoping to modernize their existing infrastructure to use more serverless offerings in order to take advantage of the cloud. Because of the timing of their contract renewal with the colocation facility, they have only 2 months for their initial migration. How would you recommend they approach their upcoming migration strategy so they can maximize their cost savings in the cloud while still executing the migration in time?

- A. Migrate the workloads to Dataproc plus HDFS; modernize later.
- B. Migrate the workloads to Dataproc plus Cloud Storage; modernize later. **Most Voted**
- C. Migrate the Spark workload to Dataproc plus HDFS, and modernize the Hive workload for BigQuery.
- D. Modernize the Spark workload for Dataflow and the Hive workload for BigQuery.

Hide Answer

Suggested Answer: D 

Community vote distribution

B (87%)

13%

by  AWSandeeep at Sept. 2, 2022, 6:57 p.m.



EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 124 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 124

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You are designing a cloud-native historical data processing system to meet the following conditions:

- ☞ The data being analyzed is in CSV, Avro, and PDF formats and will be accessed by multiple analysis tools including Cloud Dataproc, BigQuery, and Compute Engine.
- ☞ A streaming data pipeline stores new data daily.
- ☞ Performance is not a factor in the solution.
- ☞ The solution design should maximize availability.

How should you design data storage for this solution?

- A. Create a Cloud Dataproc cluster with high availability. Store the data in HDFS, and perform analysis as needed.
- B. Store the data in BigQuery. Access the data using the BigQuery Connector on Cloud Dataproc and Compute Engine.
- C. Store the data in a regional Cloud Storage bucket. Access the bucket directly using Cloud Dataproc, BigQuery, and Compute Engine.
- D. Store the data in a multi-regional Cloud Storage bucket. Access the data directly using Cloud Dataproc, BigQuery, and Compute Engine.

Hide Answer

Suggested Answer: C 🗳️

Community vote distribution

D (80%)

B (20%)

by [jvg637](#) at March 18, 2020, 8:30 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 116 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 116

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

A shipping company has live package-tracking data that is sent to an Apache Kafka stream in real time. This is then loaded into BigQuery. Analysts in your company want to query the tracking data in BigQuery to analyze geospatial trends in the lifecycle of a package. The table was originally created with ingest-date partitioning. Over time, the query processing time has increased. You need to implement a change that would improve query performance in BigQuery. What should you do?

- A. Implement clustering in BigQuery on the ingest date column.
- B. Implement clustering in BigQuery on the package-tracking ID column. **Most Voted**
- C. Tier older data onto Cloud Storage files, and leverage extended tables.
- D. Re-create the table using data partitioning on the package delivery date.

Hide Answer

Suggested Answer: A 

Community vote distribution

B (100%)

by  rickywck at March 17, 2020, 12:32 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 158 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 158

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You need to deploy additional dependencies to all nodes of a Cloud Dataproc cluster at startup using an existing initialization action. Company security policies require that Cloud Dataproc nodes do not have access to the Internet so public initialization actions cannot fetch resources. What should you do?

- A. Deploy the Cloud SQL Proxy on the Cloud Dataproc master
- B. Use an SSH tunnel to give the Cloud Dataproc cluster access to the Internet
- C. Copy all dependencies to a Cloud Storage bucket within your VPC security perimeter
- D. Use Resource Manager to add the service account used by the Cloud Dataproc cluster to the Network User role

Hide Answer

Suggested Answer: D 

Community vote distribution

C (100%)

by  rickywck at March 18, 2020, 1:46 a.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 153 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 153

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You operate an IoT pipeline built around Apache Kafka that normally receives around 5000 messages per second. You want to use Google Cloud Platform to create an alert as soon as the moving average over 1 hour drops below 4000 messages per second. What should you do?

- A. Consume the stream of data in Dataflow using Kafka IO. Set a sliding time window of 1 hour every 5 minutes. Compute the average when the window closes, and send an alert if the average is less than 4000 messages. **Most Voted**
- B. Consume the stream of data in Dataflow using Kafka IO. Set a fixed time window of 1 hour. Compute the average when the window closes, and send an alert if the average is less than 4000 messages.
- C. Use Kafka Connect to link your Kafka message queue to Pub/Sub. Use a Dataflow template to write your messages from Pub/Sub to Bigtable. Use Cloud Scheduler to run a script every hour that counts the number of rows created in Bigtable in the last hour. If that number falls below 4000, send an alert.
- D. Use Kafka Connect to link your Kafka message queue to Pub/Sub. Use a Dataflow template to write your messages from Pub/Sub to BigQuery. Use Cloud Scheduler to run a script every five minutes that counts the number of rows created in BigQuery in the last hour. If that number falls below 4000, send an alert.

Hide Answer

Suggested Answer: C 

Community vote distribution

A (100%)

by [deleted] at March 22, 2020, 8:12 a.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 185 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 185

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You issue a new batch job to Dataflow. The job starts successfully, processes a few elements, and then suddenly fails and shuts down. You navigate to the Dataflow monitoring interface where you find errors related to a particular DoFn in your pipeline. What is the most likely cause of the errors?

- A. Job validation
- B. Exceptions in worker code **Most Voted**
- C. Graph or pipeline construction
- D. Insufficient permissions

Hide Answer

Suggested Answer: D 

Community vote distribution

B (100%)

by  AWSandeep at Sept. 2, 2022, 10:47 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 93 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 93

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You're using Bigtable for a real-time application, and you have a heavy load that is a mix of read and writes. You've recently identified an additional use case and need to perform hourly an analytical job to calculate certain statistics across the whole database. You need to ensure both the reliability of your production application as well as the analytical workload.

What should you do?

- A. Export Bigtable dump to GCS and run your analytical job on top of the exported files. profile for the analytics workload. profile for the analytics workload.
- D. Increase the size of your existing cluster twice and execute your analytics workload on your new resized cluster.

Hide Answer

Suggested Answer: *B* 

by  [madhu1171](#) at March 15, 2020, 2:30 a.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 183 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 183

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You are using Bigtable to persist and serve stock market data for each of the major indices. To serve the trading application, you need to access only the most recent stock prices that are streaming in. How should you design your row key and tables to ensure that you can access the data with the simplest query?

- A. Create one unique table for all of the indices, and then use the index and timestamp as the row key design. **Most Voted**
- B. Create one unique table for all of the indices, and then use a reverse timestamp as the row key design. **Most Voted**
- C. For each index, have a separate table and use a timestamp as the row key design.
- D. For each index, have a separate table and use a reverse timestamp as the row key design.

Hide Answer

Suggested Answer: D 

Community vote distribution



by  AWSandeeep at Sept. 2, 2022, 10:25 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 100 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 100

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You have a requirement to insert minute-resolution data from 50,000 sensors into a BigQuery table. You expect significant growth in data volume and need the data to be available within 1 minute of ingestion for real-time analysis of aggregated trends. What should you do?

- A. Use bq load to load a batch of sensor data every 60 seconds.
- B. Use a Cloud Dataflow pipeline to stream data into the BigQuery table. **Most Voted**
- C. Use the INSERT statement to insert a batch of data every 60 seconds.
- D. Use the MERGE statement to apply updates in batch every 60 seconds.

Hide Answer

Suggested Answer: C 

Community vote distribution

B (100%)

by  jvg637 at March 19, 2020, 5:39 p.m.



EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 96 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 96

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You want to analyze hundreds of thousands of social media posts daily at the lowest cost and with the fewest steps.

You have the following requirements:

- ☞ You will batch-load the posts once per day and run them through the Cloud Natural Language API.
- ☞ You will extract topics and sentiment from the posts.
- ☞ You must store the raw posts for archiving and reprocessing.
- ☞ You will create dashboards to be shared with people both inside and outside your organization.

You need to store both the data extracted from the API to perform analysis as well as the raw social media posts for historical archiving. What should you do?

- A. Store the social media posts and the data extracted from the API in BigQuery.
- B. Store the social media posts and the data extracted from the API in Cloud SQL.
- C. Store the raw social media posts in Cloud Storage, and write the data extracted from the API into BigQuery. **Most Voted**
- D. Feed to social media posts into the API directly from the source, and write the extracted data from the API into BigQuery.

Hide Answer

Suggested Answer: C

Community vote distribution

C (100%)

by [rickywck](#) at March 17, 2020, 9:51 a.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 85 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 85

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You are a head of BI at a large enterprise company with multiple business units that each have different priorities and budgets. You use on-demand pricing for BigQuery with a quota of 2K concurrent on-demand slots per project. Users at your organization sometimes don't get slots to execute their query and you need to correct this. You'd like to avoid introducing new projects to your account.

What should you do?

- A. Convert your batch BQ queries into interactive BQ queries.
- B. Create an additional project to overcome the 2K on-demand per-project quota.
- C. Switch to flat-rate pricing and establish a hierarchical priority model for your projects.
- D. Increase the amount of concurrent slots per project at the Quotas page at the Cloud Console.

Hide Answer

Suggested Answer: C 

Reference -

<https://cloud.google.com/blog/products/gcp/busting-12-myths-about-bigquery>

Community vote distribution

C (100%)

by [deleted] at March 22, 2020, 6:02 p.m.

EXAM PROFESSIONAL MACHINE LEARNING ENGINEER TOPIC 1 QUESTION 17 DISCUSSION

Actual exam question from Google's Professional Machine Learning Engineer

Question #: 17

Topic #: 1

[\[All Professional Machine Learning Engineer Questions\]](#)

You are building a real-time prediction engine that streams files which may contain Personally Identifiable Information (PII) to Google Cloud. You want to use the Cloud Data Loss Prevention (DLP) API to scan the files. How should you ensure that the PII is not accessible by unauthorized individuals?

- A. Stream all files to Google Cloud, and then write the data to BigQuery. Periodically conduct a bulk scan of the table using the DLP API.
- B. Stream all files to Google Cloud, and write batches of the data to BigQuery. While the data is being written to BigQuery, conduct a bulk scan of the data using the DLP API.
- C. Create two buckets of data: Sensitive and Non-sensitive. Write all data to the Non-sensitive bucket. Periodically conduct a bulk scan of that bucket using the DLP API, and move the sensitive data to the Sensitive bucket.
- D. Create three buckets of data: Quarantine, Sensitive, and Non-sensitive. Write all data to the Quarantine bucket. Periodically conduct a bulk scan of that bucket using the DLP API, and move the data to either the Sensitive or Non-Sensitive bucket. **Most Voted**

Hide Answer

Suggested Answer: A 

Community vote distribution



by  [chohan](#) at June 16, 2021, 1:58 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 190 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 190

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You are loading CSV files from Cloud Storage to BigQuery. The files have known data quality issues, including mismatched data types, such as STRINGS and INT64s in the same column, and inconsistent formatting of values such as phone numbers or addresses. You need to create the data pipeline to maintain data quality and perform the required cleansing and transformation. What should you do?

- A. Use Data Fusion to transform the data before loading it into BigQuery. **Most Voted**
- B. Use Data Fusion to convert the CSV files to a self-describing data format, such as AVRO, before loading the data to BigQuery.
- C. Load the CSV files into a staging table with the desired schema, perform the transformations with SQL, and then write the results to the final destination table.
- D. Create a table with the desired schema, load the CSV files into the table, and perform the transformations in place using SQL.

Hide Answer

Suggested Answer: D 

Community vote distribution

A (100%)

by  AWSandeep at Sept. 2, 2022, 11:15 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 180 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 180

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You are migrating an application that tracks library books and information about each book, such as author or year published, from an on-premises data warehouse to BigQuery. In your current relational database, the author information is kept in a separate table and joined to the book information on a common key. Based on Google's recommended practice for schema design, how would you structure the data to ensure optimal speed of queries about the author of each book that has been borrowed?

- A. Keep the schema the same, maintain the different tables for the book and each of the attributes, and query as you are doing today.
- B. Create a table that is wide and includes a column for each attribute, including the author's first name, last name, date of birth, etc.
- C. Create a table that includes information about the books and authors, but nest the author fields inside the author column. **Most Voted**
- D. Keep the schema the same, create a view that joins all of the tables, and always query the view.

Hide Answer

Suggested Answer: D 

Community vote distribution

C (100%)

by  AWSandeep at Sept. 2, 2022, 9:22 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 192 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 192

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

You are implementing a chatbot to help an online retailer streamline their customer service. The chatbot must be able to respond to both text and voice inquiries. You are looking for a low-code or no-code option, and you want to be able to easily train the chatbot to provide answers to keywords. What should you do?

- A. Use the Cloud Speech-to-Text API to build a Python application in App Engine.
- B. Use the Cloud Speech-to-Text API to build a Python application in a Compute Engine instance.
- C. Use Dialogflow for simple queries and the Cloud Speech-to-Text API for complex queries.
- D. Use Dialogflow to implement the chatbot, defining the intents based on the most common queries collected. **Most Voted**

Hide Answer

Suggested Answer: *D* 

Community vote distribution

3 votes.

D (82%)

C (18%)

by  PhuocT at Sept. 2, 2022, 7:54 p.m.

EXAM PROFESSIONAL DATA ENGINEER TOPIC 1 QUESTION 112 DISCUSSION

Actual exam question from Google's Professional Data Engineer

Question #: 112

Topic #: 1

[\[All Professional Data Engineer Questions\]](#)

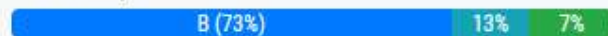
You operate a logistics company, and you want to improve event delivery reliability for vehicle-based sensors. You operate small data centers around the world to capture these events, but leased lines that provide connectivity from your event collection infrastructure to your event processing infrastructure are unreliable, with unpredictable latency. You want to address this issue in the most cost-effective way. What should you do?

- A. Deploy small Kafka clusters in your data centers to buffer events.
- B. Have the data acquisition devices publish data to Cloud Pub/Sub. **Most Voted**
- C. Establish a Cloud Interconnect between all remote data centers and Google.
- D. Write a Cloud Dataflow pipeline that aggregates all data in session windows.

Hide Answer

Suggested Answer: B 

Community vote distribution



by [deleted] at March 22, 2020, 1:58 p.m.